

EXAM MONITORING SYSTEM

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PROBLEMS

- Online exams lack monitoring capabilities and tools to identify malpractices
- Enterprise level or big proctoring tools may be impractical for day to day quizzes or exams
- heavy or complicated integration with quiz or exam platforms

FEATURES AND SOLUTIONS

- Lightweight browser based tools that integrates easily with other platforms
- Facial recognition to ensure candidate is same as the registered candidate
- Realtime monitoring that ensure continuous monitoring for malpractices
- Gives a log of all the recorded violations with the time frame

METHODOLOGY

VERIFICATION

Facial recognition using webcam compares a preregistered candidate image to a real time image

MONITORING

Mediapipe detects face in real time to ensure

- only one candidate is in the frame at time
- Candidate is present in the frame at all times

LOGGING

Generates a CSV report with

- time stamped violations
- violation type

TECH STACK

Libraries & Tools:

- Streamlit: Frontend deployment
- DeepFace: Facial recognition (Model VGG-Face)
- MediaPipe: Face detection
- OpenCV: Image processing

CHALLENGES

- **Feature-Speed Trade-offs:** Scaled back advanced features (e.g., head position tracking) to prioritize core accuracy and real-time performance.
- **Dependency Conflicts:** Debugged compatibility issues between OpenCV, DeepFace, and Streamlit.
- **Resource Limits:** Optimized frame processing for low-end device

FUTURE ENHANCEMENTS

- **Interval-Based Logging**

Log violations periodically (e.g., every 2 mins) to reduce noise.

- **Frame-Skipping Monitoring**

Process every 3rd/5th frame to lower CPU/GPU usage.

- **Lightweight Model Integration**

Test faster models (YOLO, MobileNet) for real-time efficiency.

The background features a light gray field with abstract teal geometric elements. In the top-left, there are nested rectangular outlines and a diagonal line. The top-right corner contains a 4x5 grid of small teal circles. The bottom-left has a 5x4 grid of similar circles. The bottom-right features more nested rectangular outlines. A diagonal teal line runs from the top-right towards the bottom-left, intersecting the other elements.

THANK YOU